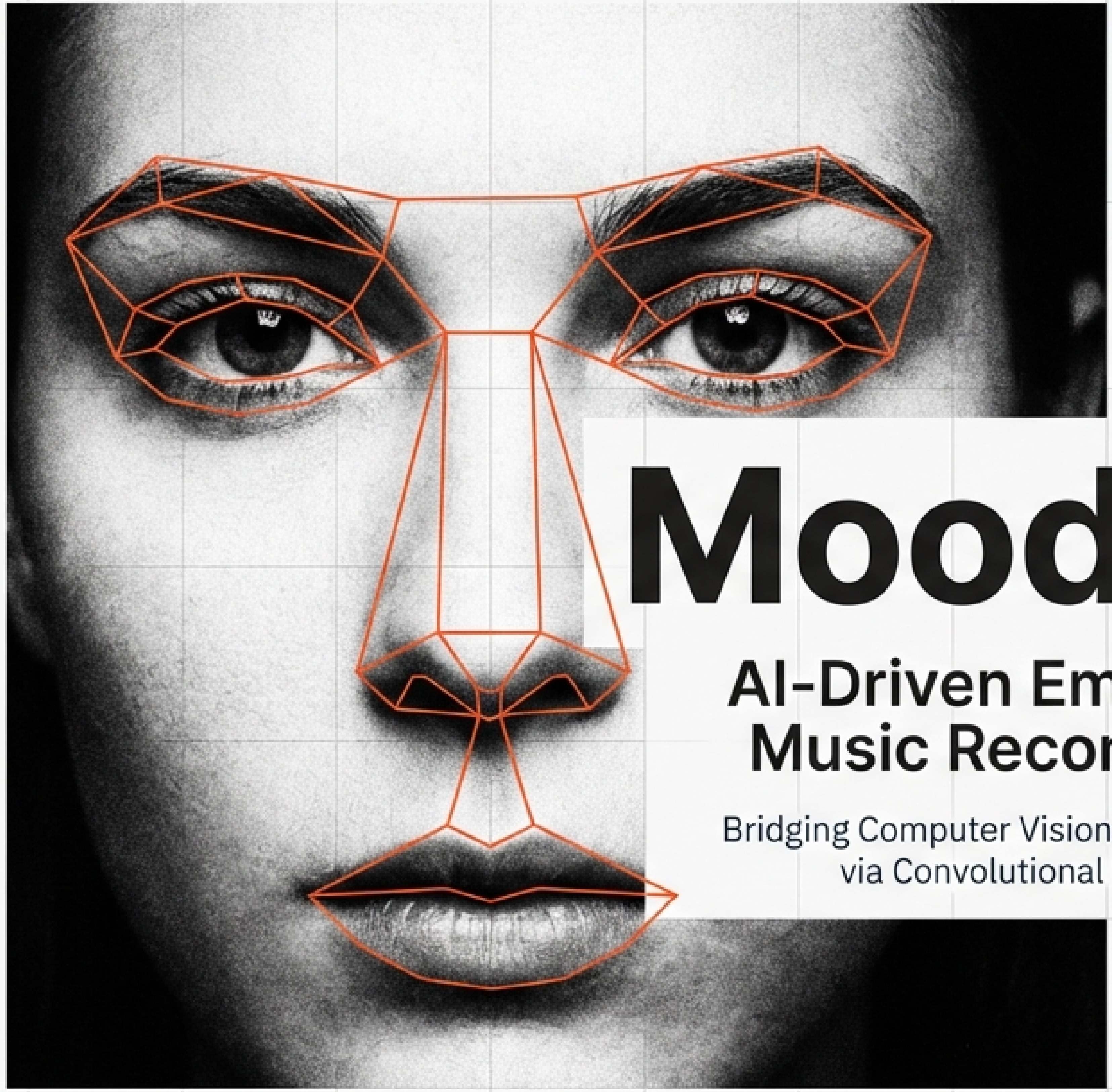


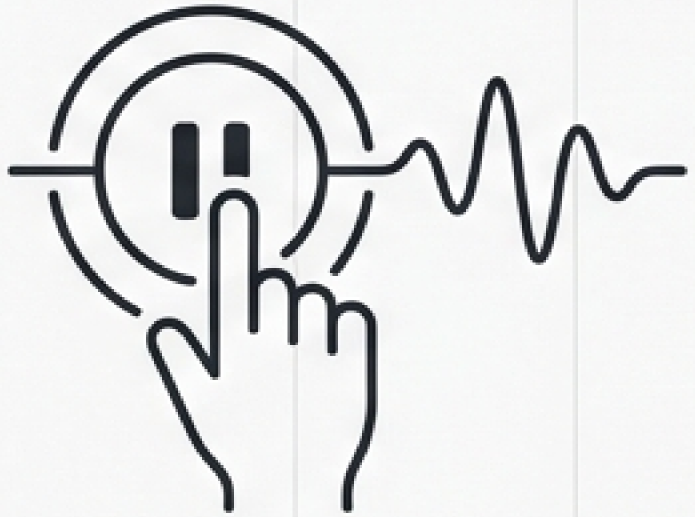
MoodMate

AI-Driven Emotion-Based Music Recommendation

Bridging Computer Vision and Auditory Experience via Convolutional Neural Networks.

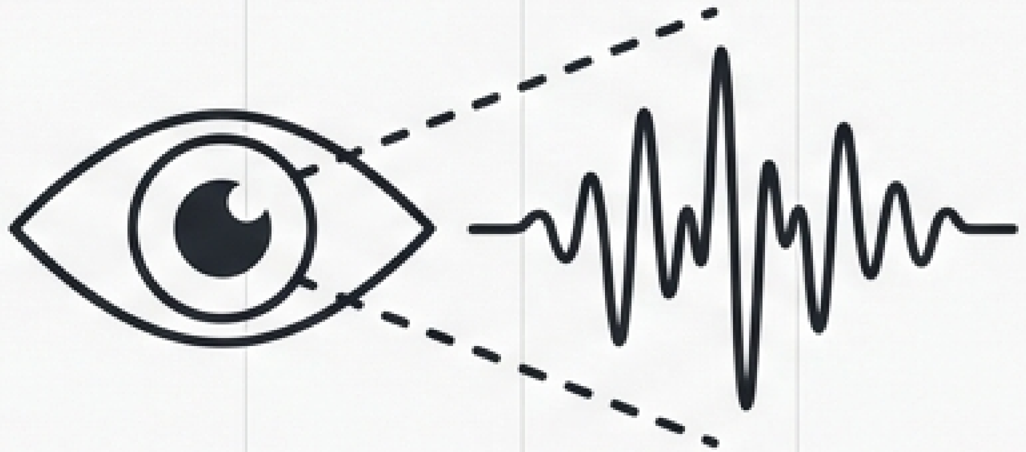
TECHNICAL PRODUCT SHOWCASE

Automating the Soundtrack to Your Life



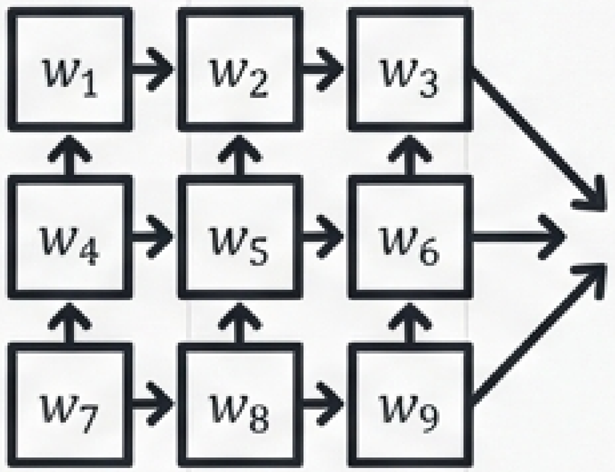
The Problem

Manual music selection requires active effort that often disrupts the listener's flow. Existing platforms rely on manual input or historical habits rather than the user's current physiological state.



The Solution

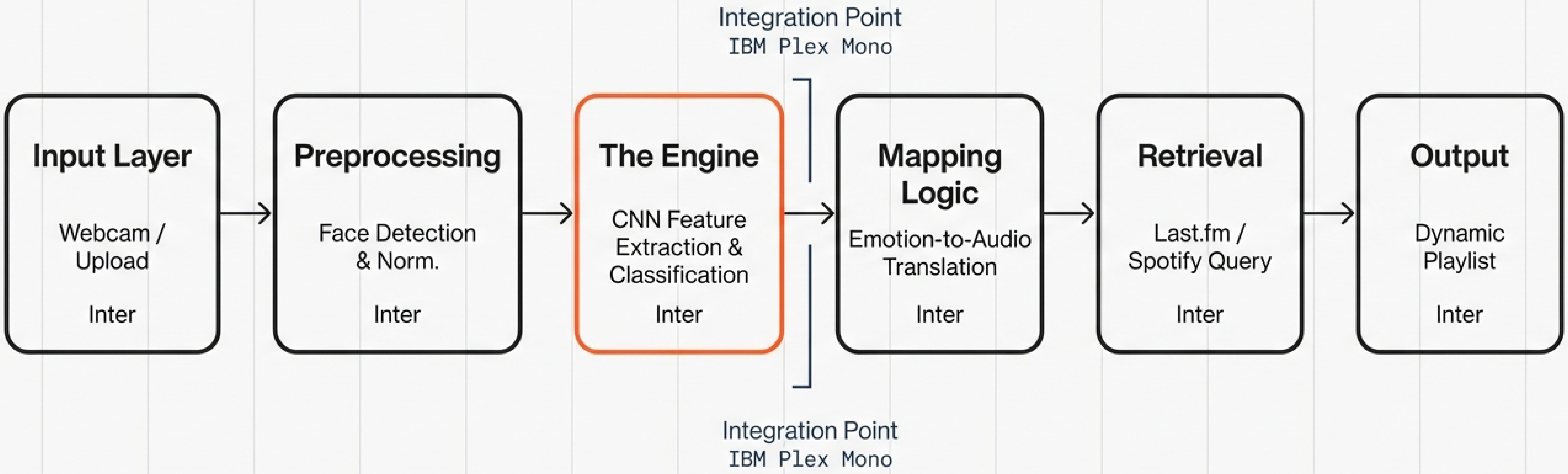
MoodMate constitutes an intelligent system that detects emotional states from facial expressions (via webcam or image upload) to recommend music that aligns with or enhances the user's mood.



Core Technology

A custom-trained **Convolutional Neural Network** (CNN) trained on the `FER-2013` dataset integrated with a logic-based recommendation engine.

End-to-End System Architecture



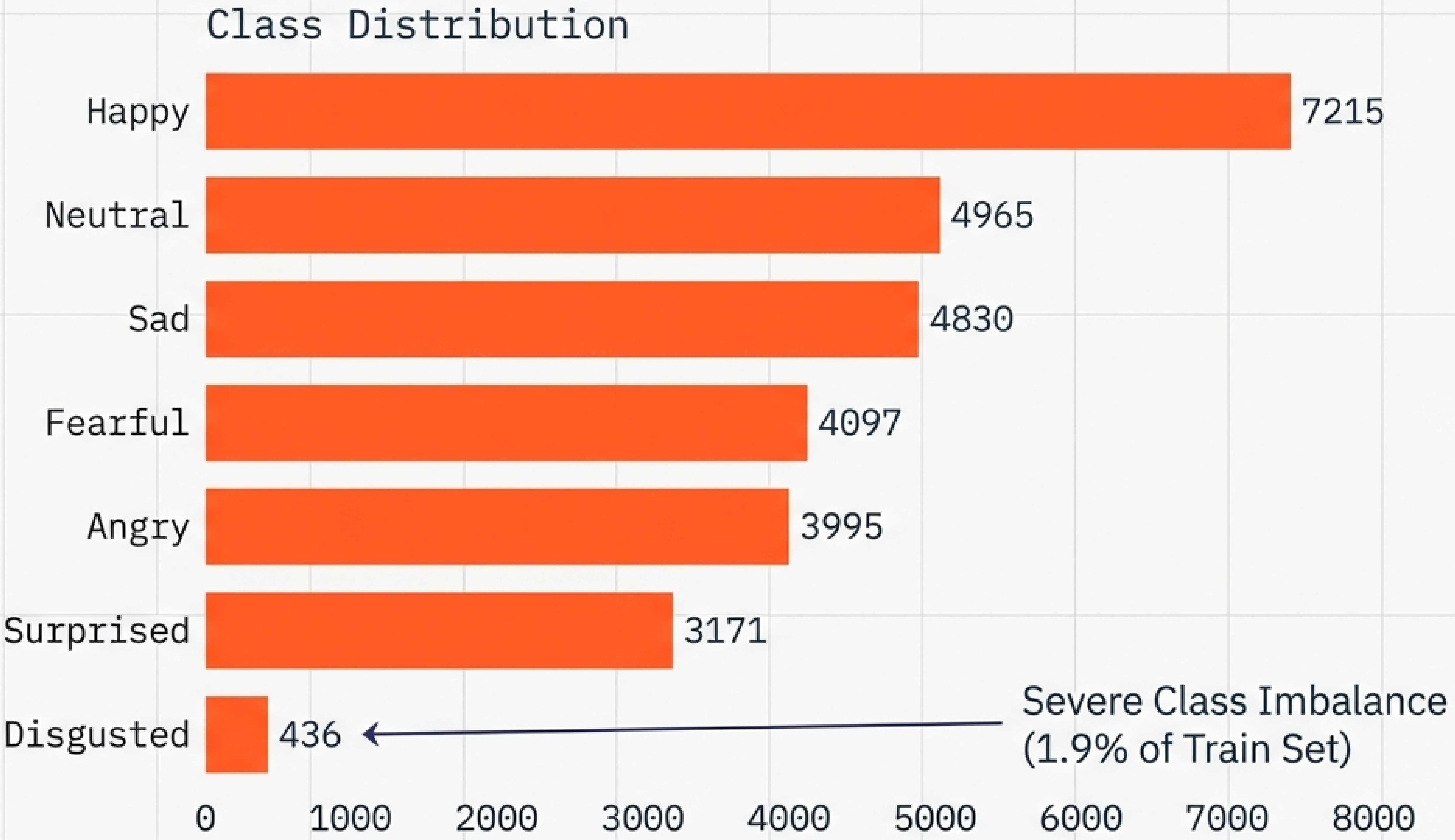
Training Foundation: The FER-2013 Dataset

Dataset Splits

Training Set: 22,968 images

Validation Set: 5,741 images

Test Set: 7,178 images

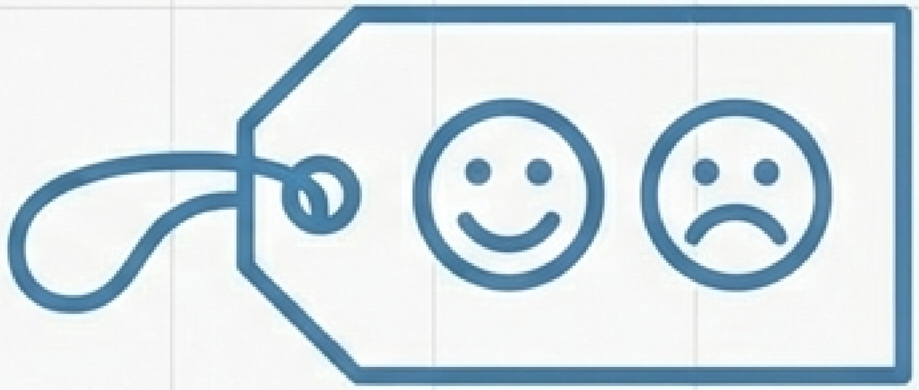


Auditory Data: Feature Extraction

Quantifying Music via Last.fm & Spotify Metadata

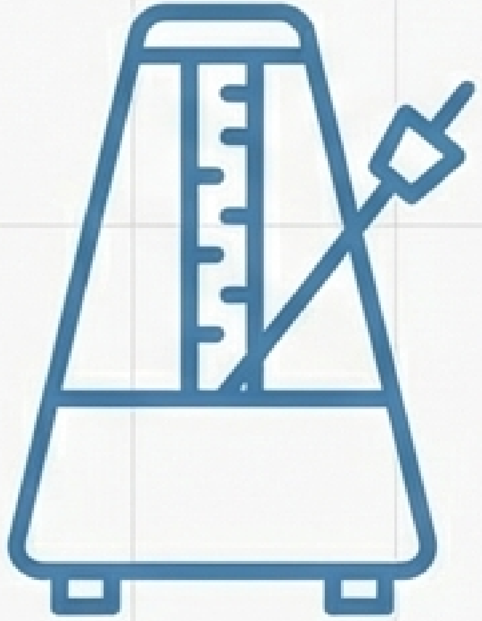
Genre

Categorical Classification



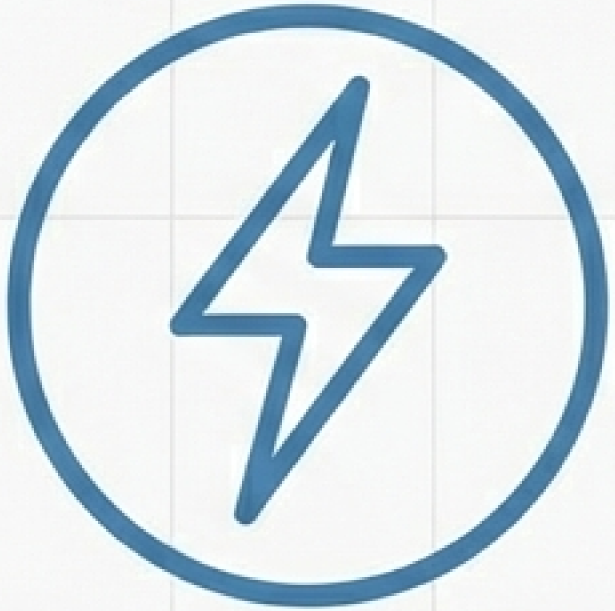
Mood Tags

Semantic Descriptors (e.g., Chill, Aggressive)



Tempo

BPM (Beats Per Minute)



Energy

Intensity & Volume Levels

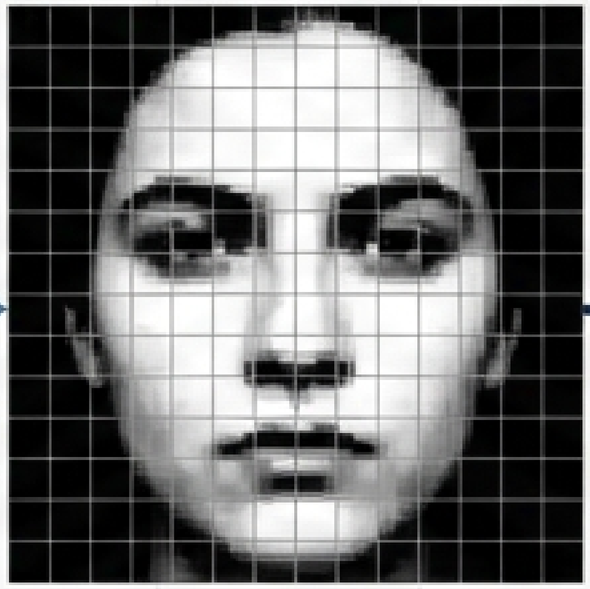
Methodology: Preprocessing & Augmentation

Raw Input

Grayscale

IBM Plex Mono
Channel Reduction

Resizing



48x48px
Normalization

Augmentation

Robustness
Training

Data preparation standardizes inputs and artificially expands the training set to prevent overfitting on limited samples.

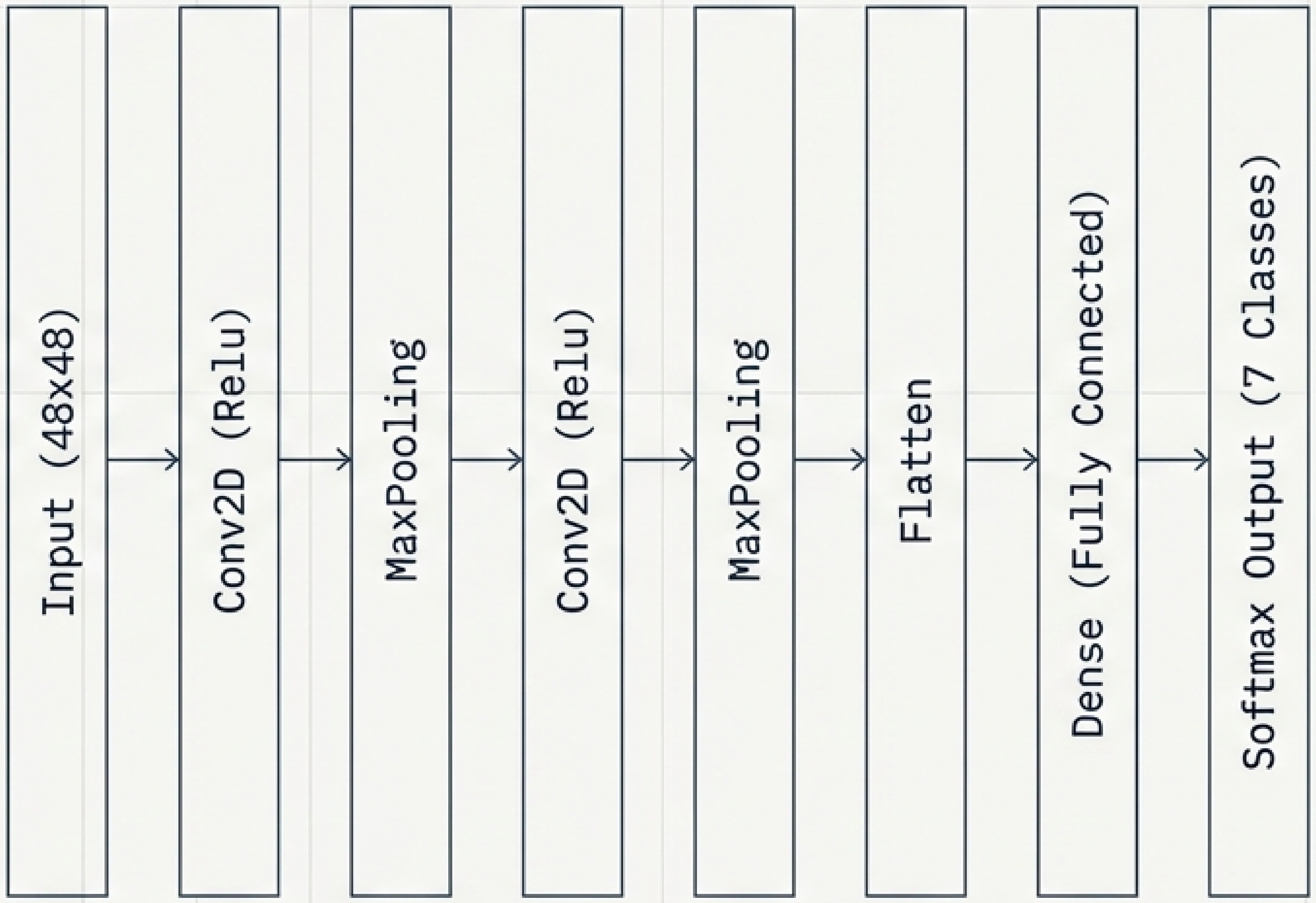
Model Architecture: Custom CNN

Architecture Design

A Convolutional Neural Network built from scratch, optimized for the specific constraints of FER-2013.

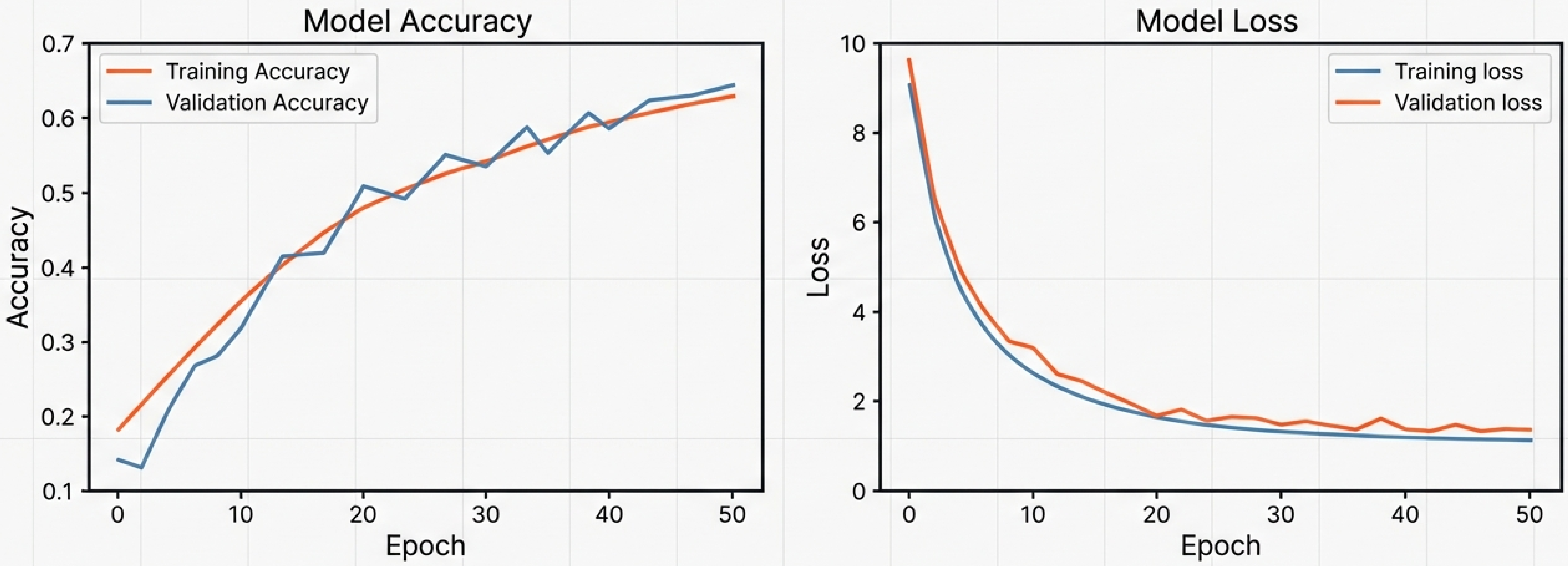
Design Philosophy

1. **Hierarchical Extraction**: Edges -> Shapes -> Facial Structures.
2. **Efficiency**. Hyperparameter tuned for real-time inference on standard consumer hardware.
3. **Customization**: Avoids bloat of pre-trained models (like ResNet) for this specific task.



Sequential Model Structure

Training Dynamics and Convergence



Interpretation: Convergence of training and validation metrics indicates stable learning with minimal overfitting.

Evaluation Metrics

Classification Report				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
Angry	0.51	0.62	0.56	958
Disgusted	0.37	0.71	0.48	111
Fearful	0.56	0.28	0.37	1024
Happy	**0.87**	**0.81**	**0.84**	1774
Neutral	0.55	0.66	0.60	1233
Sad	0.51	0.50	0.50	1247
Surprised	0.71	0.81	0.76	831
Global Accuracy: 63%				

**Analysis:

"Happy" achieves highest reliability (F1 0.84) due to ample training data.

"Fearful" suffers from low recall (0.28), often misclassified due to visual subtlety.

Analyzing Misclassification: The Confusion Matrix

	Angry	Disgusted	Fearful	Happy	Neutral	Sad	Surprised
Angry	596	51	40	35	110	97	29
Disgusted	19	79	0	0	4	7	2
Fearful	169	21	283	26	141	245	139
Happy	68	15	19	1430	140	52	50
Neutral	115	8	40	60	811	171	28
Sad	172	37	78	52	261	620	27
Surprised	32	5	49	32	20	22	671

Model struggles to distinguish between 'Sad' and 'Neutral' due to lack of distinct facial landmarks in those expressions.
Inter Regular, clean leading.

The Mapping Mechanism

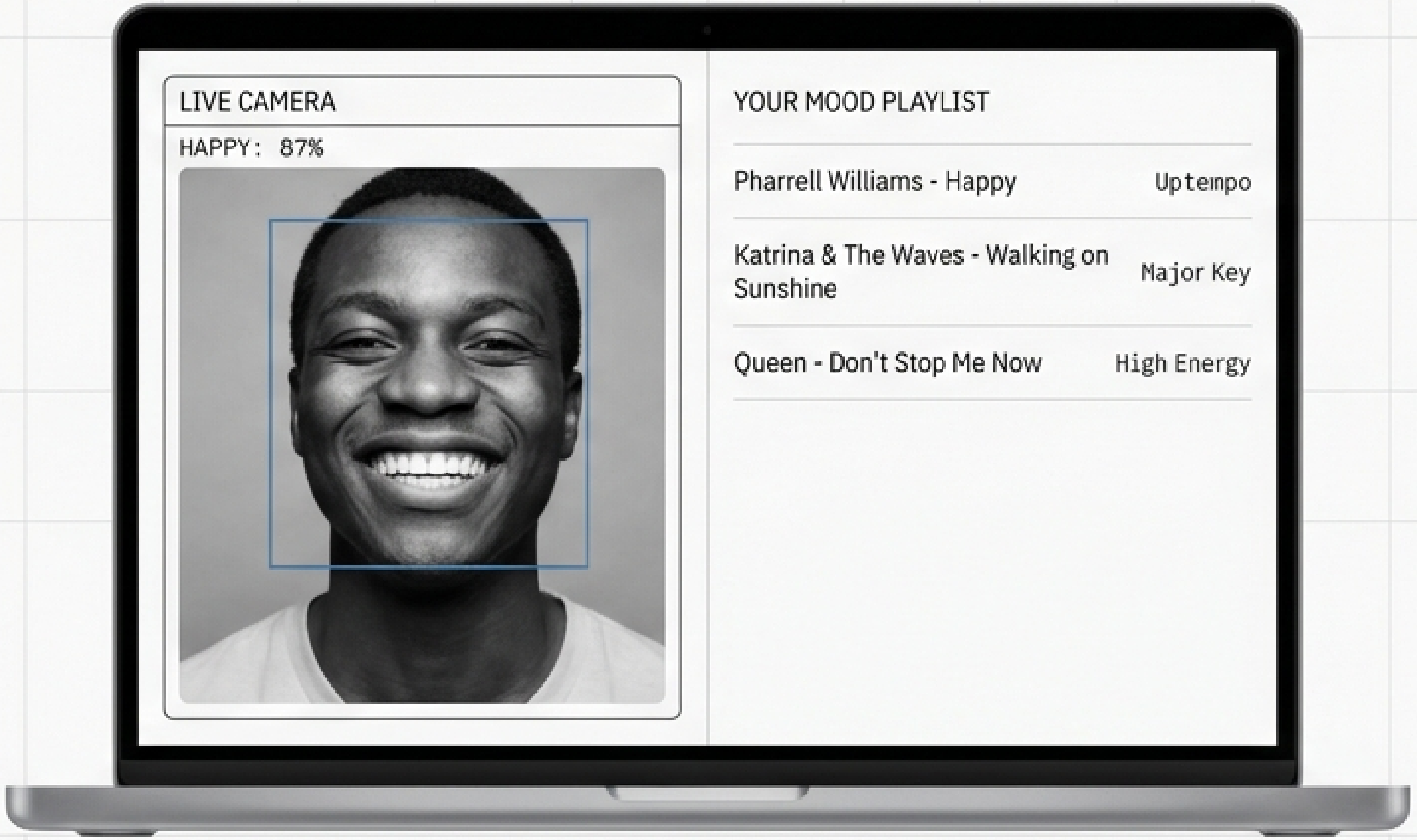
INPUT: HAPPY → Major Key + High Energy + Uptempo

INPUT: SAD → Minor Key + Low Tempo + “Melancholic” Tags

INPUT: ANGRY → High Intensity + Distorted + Rock/Metal Genre

Logic Layer → Ranked Playlist Generation

User Interface Implementation

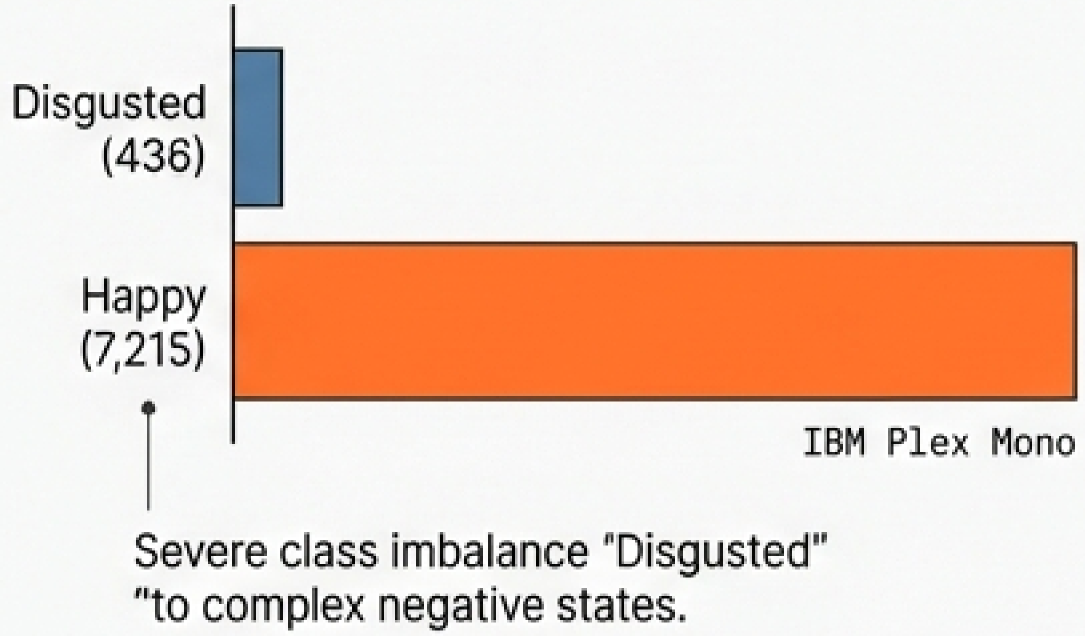


- Real-time webcam capture and inference.
- Live confidence score visualization.
- Dynamic playlist retrieval via API.

Challenges & Limitations

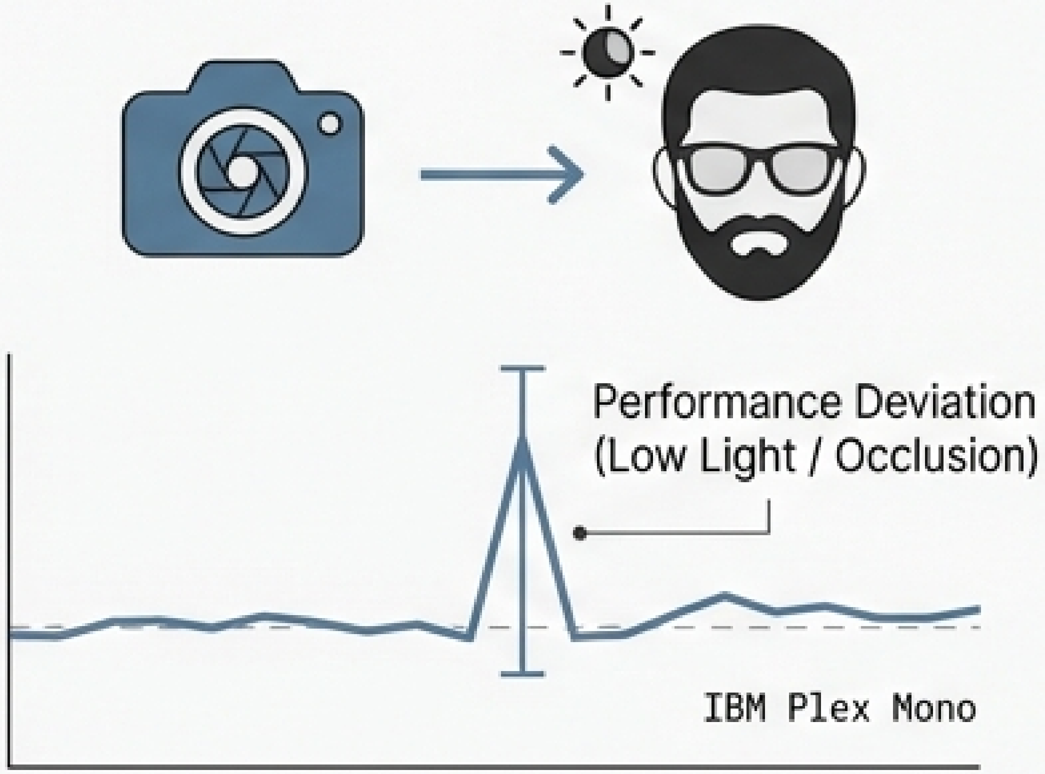
Dataset Bias

Severe class imbalance (436 “Disgusted” vs 7,215 “Happy”) creates a prediction bias toward positive emotions, reducing system sensitivity to complex negative states.



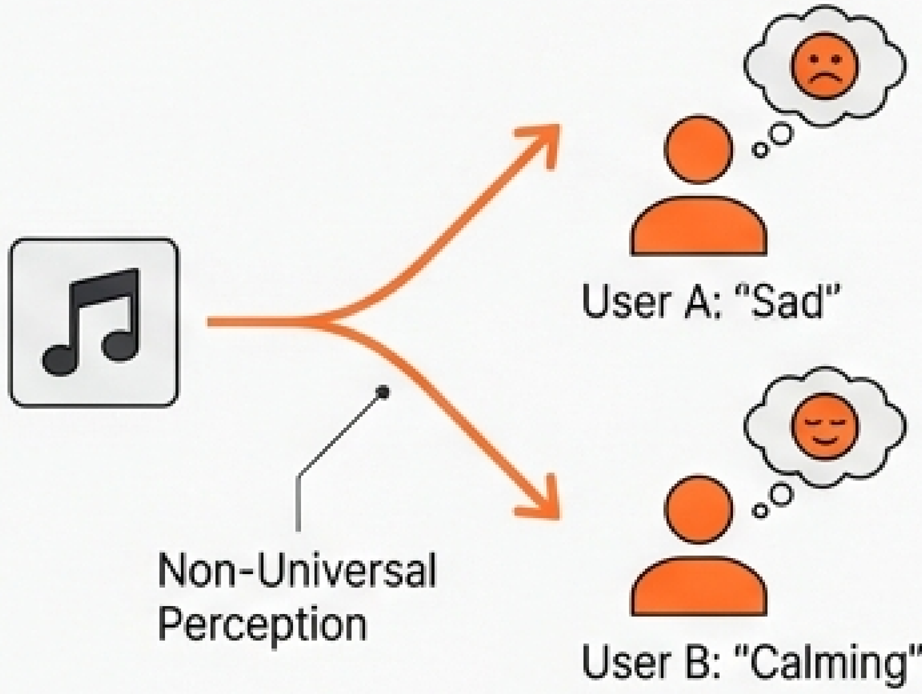
Visual Noise

Standard deviation in performance increases significantly in low-light environments or with facial occlusions (glasses, facial hair, hands).



Subjectivity

Music perception is non-universal. A “Sad” track might be perceived as “Calming” by different users, challenging the rigidity of hard-coded logic mapping.



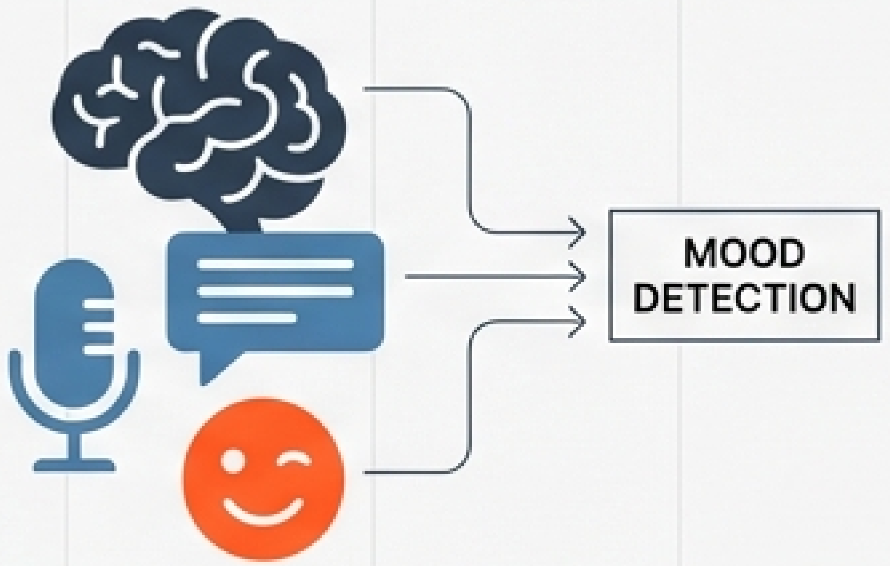
Future Scope

1 → 2 → 3

Multimodal Input

Neue Haas Grotesk Display Pro

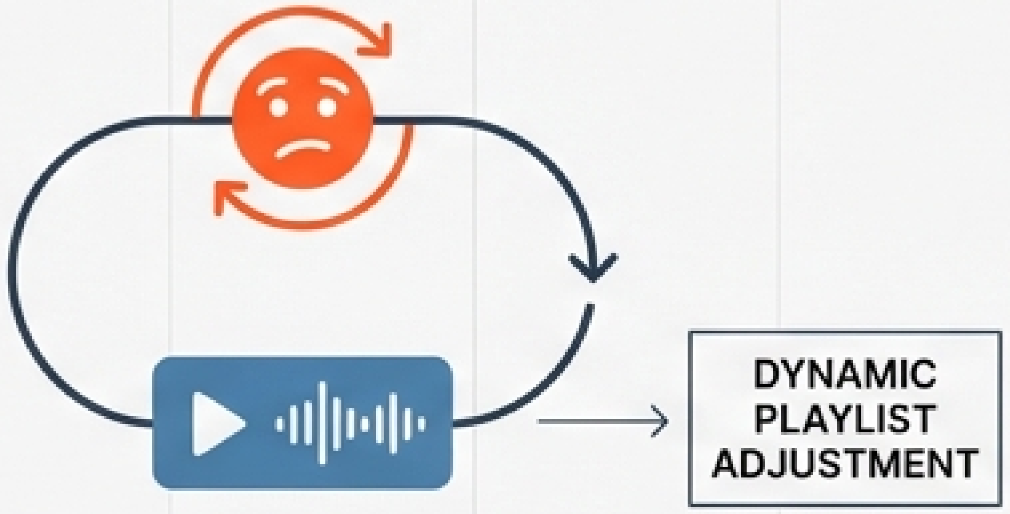
Inter: Integrating NLP (Natural Language Processing) to analyze text or voice input alongside facial expressions for higher-fidelity mood detection.



Real-Time Tracking

Neue Haas Grotesk Display Pro

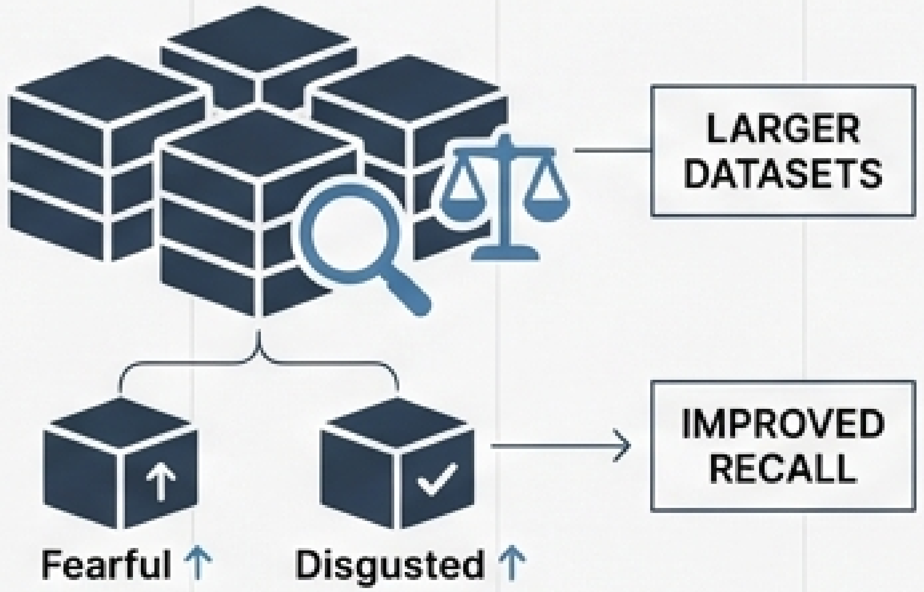
Inter. Continuous monitoring during song playback to dynamically skip or change tracks if the user's expression shifts.



Data Expansion

Neue Haas Grotesk Display Pro

Inter: Incorporating larger, balanced datasets to improve recall on under-represented classes like 'Fearful' and 'Disgusted'.



Conclusion

MoodMate successfully demonstrates an end-to-end pipeline processing 22,000+ images to deliver personalized audio experiences.

While current global accuracy stands at 63%, the system validates the core hypothesis: AI can effectively translate biological signals into aesthetic recommendations, paving the way for empathetic computing.